

Mehedi Hasan Bijoy

Speech and Language Processing Researcher

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SUMMARY

Doctoral Researcher in Speech and Language Processing, specializing in accent-robust ASR and multilingual speech systems. Skilled in designing, implementing, and deploying end-to-end machine learning pipelines, with hands-on experience in Fine-Tuning, LoRA, Knowledge Distillation, RAG, and Prompt Engineering. Authored several peer-reviewed publications in Q1 journals and top-tier conferences, evidencing a strong track record of novel, impactful, and experimentally rigorous research in speech and language technologies.

RESEARCH INTERESTS

Accented/Dialectal Speech Recognition, Multimodal Modeling, and Low-Resourced ASR

RESEARCH HIGHLIGHTS

- ★ Proposed a Self-Supervised Accent-Discovery method with Fairness-Aware Adapter Routing that outperformed zero-shot, fine-tuning, and LoRA baselines while using only ~0.7% of FT parameters.
- ★ Improved accented English ASR robustness by ~14% relative WER using a novel Multi-Teacher Knowledge Distillation framework evaluated across 30+ accents from L1 and L2 speakers.
- ★ Achieved 5–12% performance gains in multilingual speech emotion recognition for English, Finnish, and French by leveraging Language-Aware Mixture-of-Experts architecture.
- ★ Developed two large-scale Bangla corpora for grammatical error correction and punctuation restoration, yielding >10% performance gains on both tasks.

EDUCATION

Doctor of Philosophy (Ph.D.) in Electrical Engineering / CCIS / *Speech & Language Processing* Sep. 25 —
[Aalto University](#), Finland Aug. 29

- ★ **Topic:** Accented Automatic Speech Recognition

Master of Science (M.Sc.) in [Computer, Communication, and Information Sciences](#) Oct. 23 —
[Aalto University](#), Finland Jun. 25

- ★ **CGPA:** 4.83 / 5.00
- ★ **Major:** Speech and Language Technology
- ★ **Minor:** Machine Learning, Data Science, and Artificial Intelligence
- ★ **Award & Scholarship:** I. Master's Degree with Honours, II. Finland Scholarship A-100%

Bachelor of Science (B.Sc.) in [Computer Science & Engineering](#) May 17 —
[North South University](#), Bangladesh Sep. 21

- ★ **CGPA:** 3.81 / 4.00
- ★ **Specialization:** Artificial Intelligence
- ★ **Award & Scholarship:** I. Summa Cum Laude, II. Merit-Based Scholarship.

WORK EXPERIENCE

- Sep 25 — * **Doctoral Researcher**
[Automatic Speech Recognition Research Group](#), Aalto University, Finland
→ Poster: [@FCAI Day 2025](#)
- Mar 24 — Aug 25 **Research Assistant**
[Automatic Speech Recognition Research Group](#), Aalto University, Finland
→ Publications: [@ACMM/MuSe2024](#) and [@Interspeech2025](#)
- Jan 23 — Oct 23 **Lecturer**
[Department of Computer Science](#), Bangladesh University of Business and Technology
→ Delivered lectures for 5 computer science courses (Pattern Recognition, Data Structures, OOP, SPL, and Discrete Math), teaching ~400 undergraduate students.
- Mar 22 — Jun 23 **Research Assistant**
[Institute for Advanced Research](#), United International University, Bangladesh
→ Publications: [@EMNLP/BLP2023](#), [@CSL](#), and [@Neural Computing and Applications](#)
- Feb 22 — Jan 23 **Lab Instructor**
[Department of Electrical & Computer Engineering](#), North South University, Bangladesh
→ Conducted hands-on programming labs in C and C++, teaching structured programming and data structures & algorithms to ~300 undergraduate students.

ACADEMIC & COMMUNITY SERVICES

- ★ **Reviewer**, [Workshop on Bangla Language Processing](#) at EMNLP 2023 and IJCNLP-AACL 2025.
- ★ **Teaching Assistant**, Aalto University
Courses: [Speech Recognition](#) (Autumn 24, 25), [Statistical Natural Language Processing](#) (Spring 25, 26).
- ★ **Teaching Assistant**, North South University
Course: [Probability & Statistics](#) (Fall 20, Spring 21, Summer 21).

SKILLS

Programming Languages: Python, C, C++, Java

ML Libraries & Frameworks: PyTorch, HuggingFace Transformers, LangChain, Ollama

MLOps: MLflow, Docker, Airflow, GitHub Actions, AWS, Flask, FastAPI, Grafana

Data Storage & Acquisition: SQL Databases, Vector Databases, Web Scraping (Selenium, BeautifulSoup)

Developer Tools: Git, DVC, Linux, LaTeX

PROJECTS

Speech-to-Text Web Application

Developed a Whisper-powered ASR web app using FastAPI, PyTorch, and HuggingFace Transformers, with audio upload and record handling, on-device inference, and REST API integration.

RAG-based Question-Answering System

Developed a RAG pipeline using Llama, FAISS, and LangChain, enabling context-aware question answering with document embedding, chunking, retrieval, and LLM-based response generation.

PUBLICATIONS

- ★ **Conference** **Mehedi Hasan Bijoy**, Dejan Porjazovski, Tamás Grósz, and Mikko Kurimo. "Multi-Teacher Language-Aware Knowledge Distillation for Multilingual Speech Emotion Recognition." Proc. Interspeech 2025, 146-150, doi: 10.21437/Interspeech.2025-418
- ★ **Journal** **Mehedi Hasan Bijoy**, Nahid Hossain, Salekul Islam, and Swakkhar Shatabda. "A transformer-based spelling error correction framework for Bangla and resource scarce Indic languages." Computer Speech & Language 89 (2025): 101703.
- ★ **Workshop** **Mehedi Hasan Bijoy**, Dejan Porjazovski, Nhan Phan, Guangpu Huang, Tamás Grósz, and Mikko Kurimo. "Multimodal Humor Detection and Social Perception Prediction." In Proceedings of the 5th on Multimodal Sentiment Analysis Challenge and Workshop: Social Perception and Humor, pp. 60-64. 2024.
- ★ **Journal** Nahid Hossain*, **Mehedi Hasan Bijoy***, Salekul Islam, and Swakkhar Shatabda. "Panini: a transformer-based grammatical error correction method for Bangla." Neural Computing and Applications 36, no. 7 (2024): 3463-3477.
- ★ **Journal** **Mehedi Hasan Bijoy**, Nirob Hasan, Mithun Biswas, Suvodeep Mazumdar, Andrea Jimenez, Faisal Ahmed, Mirza Rasheduzzaman, and Sifat Momen. "Towards sustainable agriculture: a novel approach for rice leaf disease detection using dCNN and enhanced dataset." IEEE Access 12 (2024): 34174-34191.
- ★ **Workshop** **Mehedi Hasan Bijoy**, Mir Fatema Afroz Faria, Mahbub E. Sobhani, Tanzid Ferdoush, and Swakkhar Shatabda. "Advancing bangla punctuation restoration by a monolingual transformer-based method and a large-scale corpus." In Proceedings of the First Workshop on Bangla Language Processing (BLP-2023), pp. 18-25. 2023.
- ★ **Conference** **Mehedi Hasan Bijoy**, Nirob Hasan, Md Tahrim Faroque Tushar, and Shafin Rahman. "Image tagging by fine-tuning class semantics using text data from web scraping." In 2021 24th International Conference on Computer and Information Technology (ICCIT), pp. 1-6. IEEE, 2021.

POSTERS & PRESENTATIONS

- ★ Multi-Teacher Language-Aware Knowledge Distillation for Multilingual Speech Emotion Recognition
Poster Presentation at FCAI Day 2025 ([Poster Link](#))
- ★ Multi-Teacher Knowledge Distillation for Accented English Speech Recognition
MSc Thesis Presentation at Aalto University ([Presentation Link](#))
- ★ Advancing Bangla Punctuation Restoration by a Monolingual Transformer-Based Method and a Large-Scale Corpus
Bangla Language Processing Workshop at EMNLP 2023 ([Presentation Link](#))